

# VPA IGCSE ICT Year 10 Semester 2 — Weeks 1–9 Consolidation Practice

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Victoria Park Academy | IGCSE ICT (0417) | Weeks 1–9

Mixed practical tasks covering all IGCSE ICT skills areas. Each week includes a timed consolidation task, peer assessment opportunity, and differentiation by level. This block ensures students revisit and refine every practical skill before mock examinations begin.

### Week 1 — Document Production Consolidation

Task 1A: Business Letter with Mail Merge

☐ Time: 45 minutes

Scenario: You work for VPA Educational Services. You need to send a personalised letter to 50 parents inviting them to the annual ICT showcase. The data source is provided as a CSV file with parent names, addresses, and student names.

Requirements:

Create a professional business letter template with appropriate letterhead

Insert merge fields for: parent name, address, student name, and date

Include a table showing the event schedule (date, time, activity, location)

Apply consistent formatting: serif font for body, sans-serif for headings

Add a footer with page number and confidentiality notice

Complete the mail merge to produce all 50 letters

Save as both .docx and .pdf formats

Bronze: Complete mail merge with 3 merge fields; basic table; standard formatting.

Silver: All Bronze + conditional formatting in table; custom letterhead design; styles applied.

Gold: All Silver + IF field to customise greeting by title (Mr/Mrs/Dr); table of contents; macro to auto-save PDF.

Task 1B: Report with Styles and Cross-References

☐ Time: 45 minutes

Scenario: Create a 5-page report on "The Impact of Artificial Intelligence on Education" for the school governors.

Requirements:

Apply heading styles (Heading 1, Heading 2, Heading 3) throughout

Insert automatic table of contents

Include at least 3 cross-references (e.g., "See Section 2.3 on page 4")

Insert 2 captioned figures with automatic figure numbering

Add footnotes for 3 key terms

Format with 2.5cm margins, 1.5 line spacing, justified text

Include header with report title and footer with page numbers

Bronze: Basic heading styles; manual table of contents; 1 captioned figure.

Silver: Auto table of contents; 2 cross-references; 2 captioned figures with auto-numbering.

Gold: All Silver + automatic cross-references; footnotes; index of key terms; professional cover page.

Peer Assessment (15 minutes): Swap documents with a partner. Use the checklist:

- ☐ Are all merge fields correctly populated?
- ☐ Is the formatting consistent throughout?
- ☐ Does the table of contents update correctly?
- ☐ Are cross-references accurate?
- ☐ Is the document professionally presented?

Give one strength and one improvement suggestion.

## Week 2 — Data Manipulation Consolidation

Task 2A: Student Gradebook with Advanced Formulas

☐ Time: 50 minutes

Scenario: You are the ICT teacher. Create a spreadsheet to track 30 students' marks across 5 assignments, calculate weighted averages, and generate performance analysis.

Requirements:

Data: 30 student names, 5 assignment scores (each out of 100), assignment weights (10%, 15%, 20%, 25%, 30%)

Calculate weighted total for each student using SUMPRODUCT or manual weighting

Assign letter grades using nested IF or VLOOKUP: A ( $\geq 90$ ), B ( $\geq 80$ ), C ( $\geq 70$ ), D ( $\geq 60$ ), F ( $< 60$ )

Calculate class average, highest score, lowest score, and standard deviation

Apply conditional formatting: green for A, yellow for C, red for F

Create a bar chart showing average score per assignment

Create a pie chart showing grade distribution

Protect the formula cells; allow data entry only in score cells

Bronze: Basic formulas; simple IF for grades; one chart; basic conditional formatting.

Silver: All Bronze + VLOOKUP for grades; two charts; data validation; cell protection.

Gold: All Silver + SUMPRODUCT; standard deviation; trend line on bar chart; named ranges; custom number format.

Task 2B: Financial Model with What-If Analysis

☐ Time: 40 minutes

Scenario: Create a financial model for a school fundraising event. Calculate break-even point and explore scenarios.

Requirements:

Inputs: ticket price, expected attendance, venue cost, catering cost per person, marketing budget

Calculate total revenue, total costs, and profit/loss

Use Goal Seek to find the ticket price needed to break even with 200 attendees

Create a data table showing profit at different attendance levels (100, 150, 200, 250, 300)

Add scroll bars or spinners to adjust ticket price and see live profit update

Format currency values with ☐ symbol and 2 decimal places

Bronze: Basic revenue/cost/profit formulas; manual scenario testing.

Silver: All Bronze + Goal Seek; one data table; currency formatting.

Gold: All Silver + two-way data table; form controls (spinners); scenario manager with 3 saved scenarios.

Peer Assessment (15 minutes): Verify your partner's formulas:

- ☐ Do the weighted averages calculate correctly? (Check with calculator)
- ☐ Does conditional formatting apply to the correct ranges?
- ☐ Are charts correctly labelled with titles and axis labels?
- ☐ Does Goal Seek produce a reasonable break-even price?

## Week 3 — Presentation Consolidation

Task 3A: Interactive Product Launch Presentation

- ☐ Time: 50 minutes

Scenario: You are presenting a new educational app "VPA Learn" to potential investors. Create a 10-slide presentation.

Requirements:

Slide 1: Title slide with professional design and presenter details

Slide 2: Problem statement — use SmartArt or diagram

Slide 3: Solution overview — product screenshot placeholder

Slide 4: Market analysis — chart showing target demographics

Slide 5: Features — use icons and bullet points with animation

Slide 6: Technology stack — diagram showing app architecture

Slide 7: Business model — table with revenue projections

Slide 8: Timeline — Gantt chart or roadmap graphic

Slide 9: Team — photos and bios layout

Slide 10: Call to action — contact information and QR code placeholder

Apply consistent master slide with logo, colour scheme, and fonts

Add transitions between slides (subtle, not distracting)

Add animations to key elements (appear, fade, or wipe only)

Include speaker notes for 3 key slides

Bronze: 8 slides with basic content; simple design; 1 chart; basic transitions.

Silver: All 10 slides; consistent master slide; 2 charts; appropriate animations; speaker notes.

Gold: All Silver + custom theme; embedded video; hyperlinks to appendix slides; rehearsed timing (use Rehearse Timings).

Peer Assessment (10 minutes): Present 3 slides to your partner. They assess:

- ☐ Is the message clear and persuasive?
- ☐ Are visuals professional and relevant?
- ☐ Do animations enhance rather than distract?
- ☐ Is the design consistent throughout?

## Week 4 — Data Analysis Consolidation

#### Task 4A: School Survey Data Analysis

☐ Time: 50 minutes

Scenario: Analyse survey data from 200 students about their ICT usage habits. The data is provided as a raw CSV file.

Requirements:

Import the CSV file into a spreadsheet; clean the data (remove duplicates, fix errors)

Create a summary table showing: average screen time per year group, most popular app by gender, percentage using devices for homework

Use pivot tables to analyse: device type by age group; app category by gender

Create a scatter graph showing relationship between screen time and self-reported productivity (1–10 scale)

Add trend line and display  $R^2$  value

Create a dashboard-style layout with 4 key charts on one page

Write a 150-word summary of findings in a text box

Bronze: Import data; basic summary table; one pivot table; one chart.

Silver: All Bronze + data cleaning; two pivot tables; scatter graph with trend line; dashboard layout.

Gold: All Silver + calculated fields in pivot table; slicers for filtering; correlation analysis; professional dashboard with consistent colours.

Peer Assessment (15 minutes): Evaluate your partner's analysis:

- ☐ Is the data clean and accurate?
- ☐ Do pivot tables answer meaningful questions?
- ☐ Are charts correctly interpreted?
- ☐ Is the written summary supported by the data?

## Week 5 — Web Authoring Consolidation

#### Task 5A: Multi-Page School Website

☐ Time: 60 minutes

Scenario: Create a 4-page website for the VPA ICT Department.

Requirements:

Page 1 (index.html): Home page with welcome message, department photo placeholder, navigation menu, and footer

Page 2 (courses.html): Table showing ICT courses by year group with descriptions and entry requirements

Page 3 (projects.html): Gallery of student projects with thumbnails, titles, and brief descriptions (use CSS grid or flexbox)

Page 4 (contact.html): Contact form with fields for name, email, subject, message; form validation using HTML5 attributes

External CSS file (style.css) with: colour scheme (#1a5276 primary, #2874a6 secondary), responsive navigation, consistent typography, footer styling

All pages must link to each other via navigation menu

Include at least one embedded image with alt text

Validate HTML using W3C validator (screenshot required)

Bronze: 3 pages with basic HTML; internal CSS; simple table; working links.

Silver: All 4 pages; external CSS; responsive navigation; contact form with validation; embedded image.

Gold: All Silver + CSS grid for gallery; media query for mobile responsiveness; custom fonts; hover effects; W3C validation passed.

Peer Assessment (15 minutes): Test your partner's website:

- ☐ Do all navigation links work?
- ☐ Does the site look consistent across pages?
- ☐ Does the contact form have validation?
- ☐ Is the HTML valid? (Check W3C screenshot)
- ☐ Is the design responsive? (Resize browser window)

## Week 6 — Database Consolidation

Task 6A: School Library Database

- ☐ Time: 60 minutes

Scenario: Design and build a relational database for the VPA school library.

Requirements:

Tables: Books (BookID, Title, Author, Genre, ISBN, YearPublished, CopiesAvailable); Members (MemberID, FirstName, LastName, YearGroup, Email); Loans (LoanID, BookID, MemberID, DateBorrowed, DateDue, DateReturned, FineAmount)

Establish relationships: Books → Loans (1:N); Members → Loans (1:N)

Enforce referential integrity with cascade updates

Create queries:

Q1: All books currently on loan (show title, member name, due date)

Q2: Overdue books (due date < today, not returned)

Q3: Most popular genre (count of loans by genre)

Q4: Members with fines > ☐0

Q5: Books published after 2020 with available copies > 0

Create a data entry form for new book records with validation rules

Create a report showing monthly loan statistics (grouped by month)

Bronze: 3 tables with relationships; 3 basic queries; simple form.

Silver: All tables with referential integrity; all 5 queries; form with validation; report with grouping.

Gold: All Silver + calculated field for fine (auto-calculate based on days overdue); parameter query for member search; subform showing loan history; professional report with logo and date.

Peer Assessment (15 minutes): Test your partner's database:

- ☐ Can you add a new book through the form?
- ☐ Do all queries return correct results?
- ☐ Does the report show accurate statistics?
- ☐ Are relationships correctly enforced? (Try deleting a book that is on loan)

## Week 7 — Mixed Practical Skills 1

Task 7: Integrated Business Project

- ☐ Time: 90 minutes

Scenario: You are the ICT coordinator for a new after-school coding club. Produce a complete set of materials to launch the club.

Part A — Document (30 min): Create a proposal letter to the Headteacher requesting funding. Include: club aims, proposed schedule, equipment list with costs (table), expected outcomes, and a chart showing projected membership growth.

Part B — Spreadsheet (30 min): Create a budget spreadsheet with: income (membership fees, sponsorships), expenses (equipment, software licences, venue hire), monthly cash flow, and break-even analysis. Use conditional formatting to highlight months with negative cash flow.

Part C — Presentation (30 min): Create a 5-slide pitch deck to attract student members. Include: what the club offers, sample projects, meeting schedule, how to join, and contact information.

Bronze: Complete all 3 parts with basic content and formatting.

Silver: All Bronze + professional formatting; working formulas; consistent design across all materials.

Gold: All Silver + mail merge for personalised letters; complex financial model; embedded video in presentation; hyperlinks between documents.

## Week 8 — Mixed Practical Skills 2

### Task 8: Integrated Data Project

☐ Time: 90 minutes

Scenario: You are analysing student performance data for the VPA academic review. Produce a complete data solution.

Part A — Database (30 min): Import student data into a database. Create tables for Students, Subjects, and Results. Establish relationships. Create a query showing each student's average grade across all subjects. Create a report showing subject performance by year group.

Part B — Spreadsheet (30 min): Export query results to a spreadsheet. Create pivot tables showing: average grade by subject and year group; grade distribution (A-F counts). Create a line chart showing grade trends over 3 years. Add a dashboard summary.

Part C — Web Page (30 min): Create a simple HTML page displaying the key findings from your analysis. Include: a summary paragraph, 2 embedded charts (as images), and a table of top-performing students. Style with CSS.

Bronze: Complete all 3 parts with basic functionality.

Silver: All Bronze + complex query with JOIN; multiple pivot tables; professional web page with external CSS.

Gold: All Silver + parameter query; interactive dashboard; responsive web page; data validation at all stages.

## Week 9 — Consolidation Review & Gap Analysis

### Task 9A: Self-Assessment Portfolio Review

☐ Time: 40 minutes

Review all work from Weeks 1–8. Complete the self-assessment grid:

| Skill Area          | Confidence (1–5) | Evidence | Target for Improvement |
|---------------------|------------------|----------|------------------------|
| Document Production |                  |          |                        |
| Data Manipulation   |                  |          |                        |
| Presentations       |                  |          |                        |
| Data Analysis       |                  |          |                        |
| Web Authoring       |                  |          |                        |
| Databases           |                  |          |                        |

### Task 9B: Personalised Revision Plan

☐ Time: 30 minutes

Based on your self-assessment, create a 4-week revision plan for the mock examination period (Weeks 10–18). For each weak area:

Identify specific skills to practise (e.g., "VLOOKUP with exact match", "CSS media queries")

Allocate time slots (minimum 30 minutes per skill)

Identify resources: textbook pages, online tutorials, past paper questions

Set measurable targets (e.g., "Complete 3 VLOOKUP tasks without help by Week 11")

Peer Assessment (20 minutes): Share your revision plan with a partner. They check:

- ☐ Is the plan realistic and specific?
- ☐ Are weak areas accurately identified?
- ☐ Are resources appropriate and accessible?
- ☐ Are targets measurable and time-bound?

Sign each other's plans as accountability partners.

Teacher Note: Collect all self-assessment grids and revision plans. Use this data to form targeted intervention groups for Weeks 10–18. Students scoring  $\leq 3$  in any area should be flagged for additional support sessions.

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